according to Einstein's special relativity laws, freely moving particles must travel along absolutely straight lines. They must maintain constant direction and constant velocity, as measured in any and every inertial reference frame. This is a fundamental tenet of special relativity.

Now, Einstein's equivalence principle guarantees that gravity cannot change this fundamental tenet of free motion: Whenever a freely moving particle, in our real, gravity-endowed Universe, enters and passes through a small, inertial (freely falling) reference frame, the particle must move along a straight line through that frame. Straight-line motion through a small inertial frame, however, is the obvious analogue of straight-line behavior as measured by surveyors in a small region of the Earth's surface; and just as such straight-line behavior in small regions on Earth implies that a line is actually a geodesic of the Earth's surface, so also the particle's straight-line motion in small regions of spacetime implies that the particle moves along a geodesic of spacetime. And what is true of this particle must be true of all particles: *Every freely moving particle (every particle on which no forces, except gravity, act) travels along a geodesic of spacetime.*

As soon as Einstein realized this, it became obvious to him that *tidal gravity is a manifestation of spacetime curvature.*

To understand why, imagine the following thought experiment (mine, not Einstein's). Stand on the ice sheet at the North Pole, holding two small balls, one in each hand (Figure 2.5). Throw the balls into the air side by side, so they rise upward along precisely parallel trajectories,and then watch them fall back to Earth. Now, in a thought experiment such as this, you can do anything you wish so long as it does not violate the laws of physics. You wish to watch the trajectories of the balls as they fall under the action of gravity, not only above the Earth's surface, but also below. For this purpose, you can pretend that the balls are made of a material that falls through the Earth's soil and rock without being slowed at all (tiny black holes would have this property), and you can pretend that you and a friend on the opposite side of the Earth, who also watches, can follow the balls' motion inside the Earth via "X-ray vision."